

Dual Switch Controlled LED Sequencer Using Arduino Uno

1. Introduction

Embedded systems play a vital role in modern electronic applications by integrating hardware and software to perform specific functions. This project focuses on designing and implementing a dual switch controlled LED sequencer using Arduino Uno. The system demonstrates how microcontrollers interact with input devices and control output peripherals in real time.

2. Problem Statement

Many embedded applications require visual indicators to represent different operational states or directions. Conventional systems require complex circuitry. This project provides a simple and programmable solution using Arduino.

3. Objective

The objective of this project is to design a bidirectional LED sequencing system controlled by switches using Arduino Uno.

4. System Architecture

The system consists of an Arduino Uno, two push-button switches as inputs, and eight LEDs as outputs. The Arduino reads switch states and generates the corresponding LED sequence.

5. Components Used

Sl. No	Component Name	Specification
1	Arduino Uno	ATmega328P
2	LEDs	Red, 5mm
3	Resistors	330Ω
4	Push Buttons	Momentary
5	Pull-down Resistors	10kΩ
6	Breadboard	Standard
7	Jumper Wires	Male-Male

6. Working Principle

When Switch 1 is pressed, LEDs glow in ascending order. When Switch 2 is pressed, LEDs glow in descending order. If no switch is pressed, all LEDs remain OFF. The Arduino continuously monitors the switch inputs and controls the LEDs accordingly.

7. Real-Time Applications

This project is used in industrial indicators, decorative lighting systems, control panels, and educational embedded training kits.

8. Advantages

- Simple and easy-to-understand design
- Low-cost implementation
- Real-time user interaction
- Scalable and flexible system
- Suitable for beginners

9. Conclusion

The dual switch controlled LED sequencer successfully demonstrates basic embedded system concepts using Arduino Uno. The project is reliable, efficient, and ideal for academic and portfolio purposes.